Gravitational Wave Speed in the Causal Diamond Framework: Automatic Consistency with GW170817

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Abstract

The multi-messenger observation of GW170817 established that the speed of gravitational waves equals the speed of light to within $|c_{gw}/c - 1| < 10^{-15}$. This single measurement eliminated a wide class of modified gravity theories, including the original TeVeS, many Horndeski models, and several scalar-tensor frameworks. Here we show that the causal diamond framework for MOND—in which the MOND acceleration constant and interpolation function are derived from the entropy structure of the observer's causal diamond in ΛCDM—predicts $c_{gw} = c$ exactly, by construction. The framework modifies the effective gravitational coupling strength $\mu(x)$ without altering the spacetime metric or the equations governing gravitational wave propagation. This is a structural feature, not a tuned parameter: because the framework is built on standard ΛCDM geometry, gravitational waves propagate on the unmodified metric and automatically satisfy the GW170817 constraint.

1. The GW170817 Constraint

On August 17, 2017, the LIGO-Virgo collaboration detected gravitational waves from a binary neutron star merger (GW170817) [1], followed 1.7 seconds later by a gamma-ray burst (GRB 170817A) detected by Fermi and INTEGRAL [2]. The near-simultaneous arrival after a ~130 million light-year journey established:

$$|c_{gw}/c - 1| < 6 \times 10^{-15} \qquad (1)$$

This is the most precise test of gravitational wave propagation speed ever performed. Its immediate consequence was devastating for modified gravity: any theory in which gravitational waves travel on a metric different from the electromagnetic metric, or in which an additional scalar or vector field modifies the effective GW speed, was ruled out unless it could produce $c_{gw} = c$ exactly [3, 4].

2. Theories Killed by GW170817

The casualty list includes:

TeVeS (Bekenstein 2004) [5]: The original tensor-vector-scalar theory for MOND predicts gravitational waves propagating on a different metric than photons, giving $c_{gw} \neq c$ in general. Ruled out in its original form [3].

Horndeski gravity (subclass): A large family of scalar-tensor theories with second-order equations of motion. Boran et al. [3] and Creminelli & Vernizzi [4] showed that GW170817 eliminates all Horndeski models with a non-minimal coupling to the Gauss-Bonnet term or a kinetic braiding term. Only the simplest subclass (conformally coupled) survives.

Massive gravity (some versions): Theories with a graviton mass predict $c_{gw} < c$. GW170817 places an upper bound on the graviton mass of $m_g < 9.5 \times 10^{-23}$ eV/c^2 [1].

Einstein-Aether and Khronometric theories: Lorentz-violating gravitational theories generically predict $c_{gw} \neq c$ unless parameters are tuned [6].

The common feature of all killed theories is that they modify the *metric* or the *propagation equation* for gravitational waves. Any theory that adds fields coupled to the metric generically alters the speed at which perturbations propagate.

3. Why the Causal Diamond Framework Survives

The causal diamond framework [7, 8, 9, 10] differs from the theories listed above in a fundamental structural way: **it does not modify the metric.**

The framework is built entirely on standard ΛCDM geometry. The Friedmann equation is unmodified. The spacetime metric is the standard FLRW metric. The causal diamond is a geometric object constructed from this metric. The MOND acceleration constant $a_0 = c^2/(2\pi r_w)$ is read off from this geometry. The interpolation function $\mu(x) = kx/(1+kx)$ is derived from the entropy structure of the diamond [8].

What the framework modifies is the *effective gravitational coupling*. In the entropy-balance picture, the fraction of degrees of freedom in the area-law (Newtonian) channel is $\mu(x)$, which is less than 1 in the low-acceleration regime. This reduces the gravitational force below Newton's prediction. But it does so without adding fields, without changing the metric, and without altering the equation governing how gravitational wave perturbations propagate.

Gravitational waves are perturbations of the metric. They propagate according to the linearized Einstein equations. In the causal diamond framework, these equations are unmodified. Therefore:

$$c_{gw} = c \quad \text{(exactly, by construction)} \qquad (2)$$

This is not a prediction that must be checked against GW170817. It is a structural feature of the framework: because the framework does not modify the metric, it cannot modify the gravitational wave speed. The GW170817 constraint is satisfied automatically.

4. Comparison with Surviving MOND Theories

Framework	Modifies metric?	c_gw = c?	Status post-GW170817
TeVeS (Bekenstein 2004)	Yes	No	Ruled out
BIMOND (Milgrom 2009)	Yes	Tunable	Constrained
AeST (Skordis & Zlosnik 2021)	Yes	Yes (designed)	Survives (built for it)
Causal diamond (this work)	No	Yes (automatic)	Survives (structural)

Table 1. MOND-compatible frameworks and their gravitational wave speed predictions. The causal diamond framework is the only one where $c_{gw} = c$ is automatic rather than designed or tuned.

The distinction between "designed" and "automatic" is worth emphasizing. AeST [11] was constructed after GW170817 specifically to produce $c_{gw} = c$. It achieves this by carefully choosing the coupling between its scalar and vector fields. This is a legitimate theoretical construction, but it required engineering the theory to satisfy the constraint. The causal diamond framework, by contrast, was not built with GW170817 in mind. It inherits $c_{gw} = c$ because it inherits the metric. There is no parameter to tune and no field coupling to choose.

5. What the Framework Does and Does Not Predict for Gravitational Waves

The framework predicts $c_{gw} = c$ exactly at all frequencies, all distances, and all gravitational field strengths. This includes the deep-MOND regime ($g << a_0$), where the effective coupling μ departs significantly from unity. Even when the gravitational force is substantially reduced by the entropy balance, the waves propagate at light speed because the metric is unmodified.

The framework does *not* predict any modification to gravitational wave amplitude or waveform in the MOND regime. In theories where the coupling changes, one might expect the strain h to be modified for sources in low-acceleration environments. The causal diamond framework has not been developed to the point where waveform predictions can be made. This would require a complete dynamical theory (analogous to AeST), which the framework does not yet provide. The present paper addresses propagation speed only.

6. Caveats

The statement $c_{gw} = c$ follows from the fact that the causal diamond framework does not modify the spacetime metric. This is a feature of the framework as currently formulated. If a future dynamical completion of the framework (e.g., a Lagrangian formulation) requires additional fields that couple to the metric, the gravitational wave speed could be affected. The present result is therefore conditional on the framework remaining metric-unmodified in its eventual dynamical form.

Additionally, the causal diamond framework is not yet a complete dynamical theory. It derives a_0 and $\mu(x)$ but does not provide field equations that can be solved for arbitrary matter configurations. The gravitational wave prediction is thus a consistency check (the framework does not contradict GW170817) rather than a dynamical prediction (the framework derives the wave equation). A complete theory is needed for waveform predictions.

7. Conclusion

The causal diamond framework for MOND predicts $c_{gw} = c$ exactly, by construction. Unlike TeVeS (ruled out), Horndeski subclasses (ruled out), and AeST (engineered to survive), the causal diamond framework inherits the unmodified ΛCDM metric and therefore cannot produce a gravitational wave speed different from light speed. The GW170817 constraint, which killed many modified gravity theories, is automatically satisfied.

This is a necessary condition for the viability of the causal diamond program, not a sufficient one. The framework must still confront other tests: the redshift evolution of a_0 [10], the performance of the entropy-derived interpolation function on galaxy data [8], and the development of a complete dynamical theory. But the most immediate existential threat to any MOND-adjacent framework—the gravitational wave speed constraint—does not apply here.

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